

**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Vlan-Trunk-OSPF-MultiVendor

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1 Langkah-Langkah Percobaan

1.1 Praktikum 1: Konfigurasi Cisco Switch

1. Masuk ke mode privileged dan mode konfigurasi global pada Cisco Switch.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
*Jun  6 03:25:30.262: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adventerprisek
9-m' passed code signing verification
```

Gambar 1: Masuk ke Mode Konfigurasi Cisco Switch

Perintah yang digunakan:

```
enable
configure terminal
```

2. Membuat VLAN 10, VLAN 20, VLAN 30, dan VLAN 40 beserta nama masing-masing VLAN.

```
Switch(config)#vlan 10
Switch(config-vlan)# name Finance
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)# name IT
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)# name Marketingg
Switch(config-vlan)#exit
Switch(config)#
```

Gambar 2: Pembuatan VLAN pada Cisco Switch

Perintah yang digunakan:

```
vlan 10
name Finance
```

```
vlan 20
name HR
```

```
vlan 30
```

```
name IT
```

```
vlan 40
```

```
name Marketing
```

3. Melakukan verifikasi VLAN yang telah dibuat menggunakan perintah *show vlan brief*.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	Finance	active	
20	HR	active	
30	IT	active	
40	Marketingg	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch#
```

Gambar 3: Verifikasi VLAN Menggunakan show vlan brief

4. Mengkonfigurasi access port untuk masing-masing VLAN sesuai topologi jaringan.

```
Switch(config)#interface gi0/2
Switch(config-if)# description PC-FINANCE-VLAN10
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/1
Switch(config-if)# description PC-HR-VLAN20
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi1/1
Switch(config-if)# description PC-IT-VLAN30
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/3
Switch(config-if)# description PC-MARKETING-VLAN40
```

Gambar 4: Konfigurasi Access Port pada Setiap VLAN

Konfigurasi yang digunakan:

```
interface gi0/2
description PC-FINANCE-VLAN10
switchport mode access
switchport access vlan 10
no shutdown
```

```

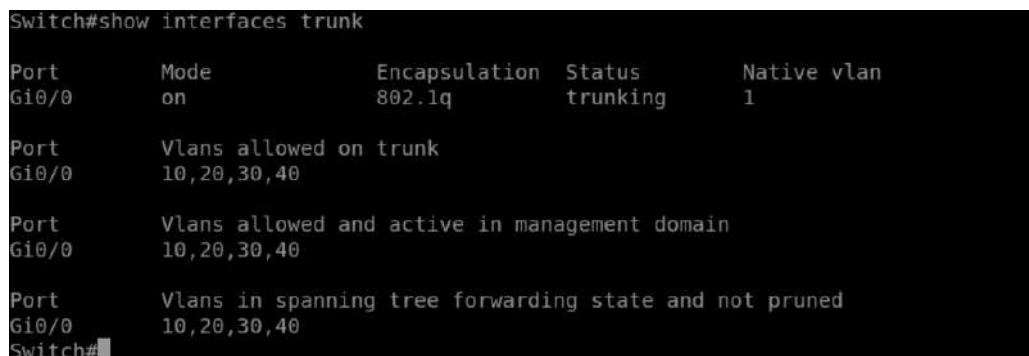
interface gi0/1
  description PC-HR-VLAN20
  switchport mode access
  switchport access vlan 20
  no shutdown

interface gi1/1
  description PC-IT-VLAN30
  switchport mode access
  switchport access vlan 30
  no shutdown

interface gi0/3
  description PC-MARKETING-VLAN40
  switchport mode access
  switchport access vlan 40
  no shutdown

```

5. Mengkonfigurasi trunk port pada interface yang terhubung dengan router Cisco.



```

Switch#show interfaces trunk

Port      Mode           Encapsulation  Status        Native vlan
Gi0/0     on             802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#

```

Gambar 5: Verifikasi Trunk Menggunakan show interfaces trunk

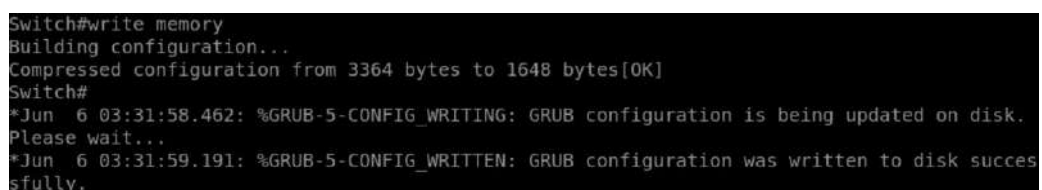
Konfigurasi yang digunakan:

```

interface gi0/0
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,40

```

6. Menyimpan konfigurasi yang telah dibuat ke memori perangkat.



```

Switch#write memory
Building configuration...
Compressed configuration from 3364 bytes to 1648 bytes[OK]
Switch#
*Jun  6 03:31:58.462: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk.
Please wait...
*Jun  6 03:31:59.191: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

```

Gambar 6: Penyimpanan Konfigurasi Cisco Switch

Perintah yang digunakan:

```
write memory
```

1.2 Praktikum 2: Konfigurasi Cisco Router

1. Mengaktifkan interface yang terhubung ke switch dan membuat subinterface VLAN 10 pada router Cisco.

```
Router(config)#interface gig0/2.10
Router(config-subif)# description GATEWAY-VLAN10-FINANCE
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)#exit
Router(config)#
```

Gambar 7: Konfigurasi Subinterface VLAN 10

```
interface GigabitEthernet0/2
no shutdown

interface GigabitEthernet0/2.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

2. Membuat subinterface VLAN 20 pada router Cisco.

```
Router(config)#interface gig0/2.20
Router(config-subif)# description GATEWAY-VLAN20-HR
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
```

Gambar 8: Konfigurasi Subinterface VLAN 20

```
interface GigabitEthernet0/2.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

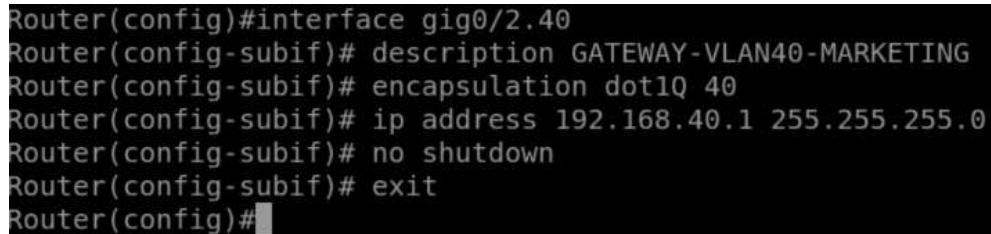
3. Membuat subinterface VLAN 30 pada router Cisco.

```
Router(config)#interface gig0/2.30
Router(config-subif)# description GATEWAY-VLAN30-IT
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

Gambar 9: Konfigurasi Subinterface VLAN 30

```
interface GigabitEthernet0/2.30
  encapsulation dot1Q 30
  ip address 192.168.30.1 255.255.255.0
```

4. Membuat subinterface VLAN 40 pada router Cisco.

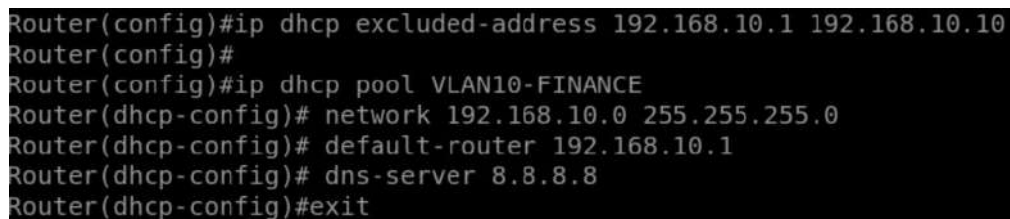


```
Router(config)#interface gig0/2.40
Router(config-subif)# description GATEWAY-VLAN40-MARKETING
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

Gambar 10: Konfigurasi Subinterface VLAN 40

```
interface GigabitEthernet0/2.40
  encapsulation dot1Q 40
  ip address 192.168.40.1 255.255.255.0
```

5. Mengkonfigurasi DHCP Server untuk VLAN 10.



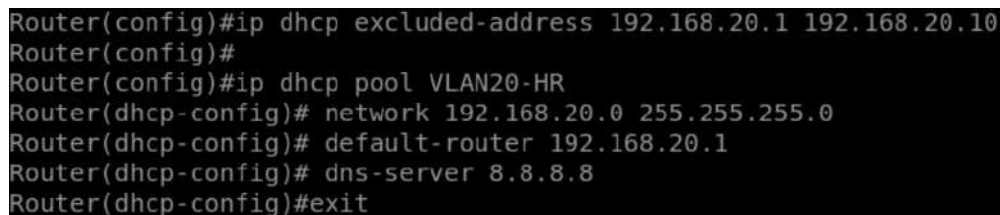
```
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
```

Gambar 11: Konfigurasi DHCP Pool VLAN 10

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

```
ip dhcp pool VLAN10-FINANCE
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 8.8.8.8
```

6. Mengkonfigurasi DHCP Server untuk VLAN 20.



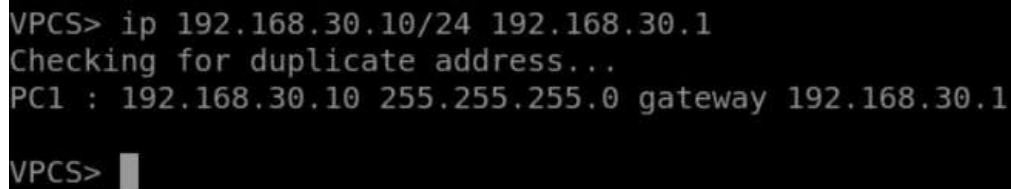
```
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
```

Gambar 12: Konfigurasi DHCP Pool VLAN 20

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10
```

```
ip dhcp pool VLAN20-HR
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
```

7. Mengkonfigurasi alamat IP statis pada PC VLAN 30.

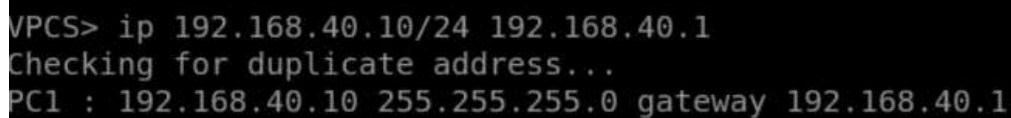


```
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1
VPCS> █
```

Gambar 13: Konfigurasi IP Statis VLAN 30

Alamat IP yang digunakan adalah 192.168.30.10/24 dengan gateway 192.168.30.1.

8. Mengkonfigurasi alamat IP statis pada PC VLAN 40.



```
VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1
```

Gambar 14: Konfigurasi IP Statis VLAN 40

Alamat IP yang digunakan adalah 192.168.40.10/24 dengan gateway 192.168.40.1.

9. Meminta alamat IP dari DHCP Server pada PC VLAN 20.



```
VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1
```

Gambar 15: Permintaan DHCP VLAN 20

Perintah yang digunakan:

```
ip dhcp
```

10. Meminta alamat IP dari DHCP Server pada PC VLAN 10.



```
VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1
```

Gambar 16: Permintaan DHCP VLAN 10

Perintah yang digunakan:

ip dhcp

11. Melakukan verifikasi konfigurasi interface router.

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES unset    administratively down down
GigabitEthernet0/1       unassigned      YES unset    administratively down down
GigabitEthernet0/2       unassigned      YES unset    up          up
GigabitEthernet0/2.10    192.168.10.1    YES manual   up          up
GigabitEthernet0/2.20    192.168.20.1    YES manual   up          up
GigabitEthernet0/2.30    192.168.30.1    YES manual   up          up
GigabitEthernet0/2.40    192.168.40.1    YES manual   up          up
GigabitEthernet0/3       unassigned      YES unset    administratively down down
GigabitEthernet0/4       unassigned      YES unset    administratively down down
GigabitEthernet0/5       unassigned      YES unset    administratively down down
GigabitEthernet0/6       unassigned      YES unset    administratively down down
```

Gambar 17: Verifikasi Interface Router

Perintah yang digunakan:

show ip interface brief

12. Melakukan verifikasi DHCP Pool yang telah dibuat.

```
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.10.12  192.168.10.1 - 192.168.10.254  1

Pool VLAN20-HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.20.12  192.168.20.1 - 192.168.20.254  1
```

Gambar 18: Verifikasi DHCP Pool

Perintah yang digunakan:

show ip dhcp pool

13. Melakukan verifikasi DHCP Binding.

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.35 Jun 07 2026 03:48 AM    Automatic
192.168.20.11    0100.5079.6668.34 Jun 07 2026 03:48 AM    Automatic
```

Gambar 19: Verifikasi DHCP Binding

Perintah yang digunakan:

```
show ip dhcp binding
```

14. Melakukan verifikasi alamat IP hasil DHCP pada VLAN 10.

```
VPCS> ip dhcp
DORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 192.168.10.11/24
GATEWAY        : 192.168.10.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.10.1
DHCP LEASE     : 86394, 86400/43200/75600
MAC            : 00:50:79:66:68:35
LPORT         : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500
```

Gambar 20: Verifikasi DHCP VLAN 10

Perintah yang digunakan:

```
show ip
```

15. Melakukan pengujian konektivitas ke gateway VLAN 10.

```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.757 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.999 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=4.440 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=4.345 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.972 ms
```

Gambar 21: Ping Gateway VLAN 10

```
ping 192.168.10.1
```

16. Melakukan verifikasi alamat IP hasil DHCP pada VLAN 20.

```

VPCS> ip dhcp
DORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 192.168.20.11/24
GATEWAY        : 192.168.20.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.20.1
DHCP LEASE     : 86396, 86400/43200/75600
MAC            : 00:50:79:66:68:34
LPORT          : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500

```

Gambar 22: Verifikasi DHCP VLAN 20

Perintah yang digunakan:

```
show ip
```

17. Melakukan pengujian konektivitas ke gateway VLAN 20.

```

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=4.708 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.600 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=4.978 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=4.815 ms

```

Gambar 23: Ping Gateway VLAN 20

```
ping 192.168.20.1
```

18. Melakukan pengujian konektivitas ke gateway VLAN 30.

```

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.845 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=4.117 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=3.824 ms

```

Gambar 24: Ping Gateway VLAN 30

```
ping 192.168.30.1
```

19. Melakukan pengujian konektivitas ke gateway VLAN 40.

```
VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.420 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=4.975 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=5.594 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=4.787 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=4.788 ms
```

Gambar 25: Ping Gateway VLAN 40

ping 192.168.40.1

20. Melakukan pengujian komunikasi antar VLAN.

```
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=4.723 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=5.717 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=6.212 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=7.725 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=5.862 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=6.296 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.625 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=6.232 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=4.202 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=5.665 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=6.686 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.119 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=4.822 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=4.790 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=4.790 ms
```

Gambar 26: Pengujian Komunikasi Antar VLAN

Pengujian dilakukan dengan melakukan ping dari VLAN 10 ke VLAN 20, VLAN 30, dan VLAN 40 untuk memastikan inter-VLAN routing berjalan dengan baik.

21. Melakukan verifikasi DHCP Binding setelah seluruh host memperoleh alamat IP.

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.35 Jun 07 2026 03:51 AM    Automatic
192.168.20.11    0100.5079.6668.34 Jun 07 2026 03:52 AM    Automatic
```

Gambar 27: Verifikasi DHCP Binding Akhir

Perintah yang digunakan:

```
show ip dhcp binding
```

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

1. Mengkonfigurasi alamat IP pada interface LAN Huawei yang terhubung ke PC LAN 50.

```
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2] description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2] ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2] undo shutdown
[*HUAWEI-Ethernet1/0/2] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 28: Konfigurasi IP LAN Huawei

Konfigurasi yang digunakan:

```
interface Ethernet1/0/2
ip address 192.168.50.1 255.255.255.0
undo shutdown
```

2. Melakukan verifikasi alamat IP pada PC LAN 50.

```
VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 192.168.50.10/24
GATEWAY        : 192.168.50.1
DNS            :
MAC            : 00:50:79:66:68:36
LPORT         : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500
```

Gambar 29: Verifikasi Alamat IP PC LAN 50

3. Melakukan pengujian konektivitas ke gateway Huawei.

```

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=10.435 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=5.208 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.702 ms

```

Gambar 30: Ping Gateway Huawei

Perintah yang digunakan:

```
ping 192.168.50.1
```

4. Mengkonfigurasi interface Cisco yang terhubung ke Huawei menggunakan jaringan 10.10.10.0/30.

```

Router(config)#interface gi0/3
Router(config-if)# description LINK1-T0-HUAWEI
Router(config-if)# ip address 10.10.10.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#

```

Gambar 31: Konfigurasi Link Cisco ke Huawei

```

interface GigabitEthernet0/3
ip address 10.10.10.1 255.255.255.252
no shutdown

```

5. Melakukan verifikasi konfigurasi interface Cisco.

```

[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0] description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0] ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0] undo shutdown
[*HUAWEI-Ethernet1/0/0] quit
[*HUAWEI]
[*HUAWEI]commit

```

Gambar 32: Verifikasi Interface Cisco

6. Mengkonfigurasi interface Cisco yang terhubung ke Huawei menggunakan jaringan 10.10.11.0/30.

```

Router(config)#interface gi0/0
Router(config-if)# description LINK2-T0-HUAWEI
Router(config-if)# ip address 10.10.11.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit

```

Gambar 33: Konfigurasi Link Kedua Cisco ke Huawei

```

interface GigabitEthernet0/0
ip address 10.10.11.1 255.255.255.252
no shutdown

```

7. Melakukan verifikasi konfigurasi interface Cisco kedua.

```
[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4] description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4] ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4] undo shutdown
[*HUAWEI-Ethernet1/0/4] quit
[*HUAWEI]
[*HUAWEI]commit
```

Gambar 34: Verifikasi Interface Cisco Kedua

8. Mengkonfigurasi link Huawei ke Mikrotik menggunakan jaringan 10.10.20.0/30.

```
[~HUAWEI]interface Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3] description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ip address 10.10.20.1 255.255.255.252
[*HUAWEI-Ethernet1/0/3] undo shutdown
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
```

Gambar 35: Konfigurasi Link Huawei ke Mikrotik

```
interface Ethernet1/0/3
ip address 10.10.20.1 255.255.255.252
undo shutdown
```

9. Melakukan verifikasi interface Mikrotik yang terhubung ke Huawei.

```
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
[admin@MikroTik] >
```

Gambar 36: Verifikasi Interface Mikrotik Ether1

10. Mengkonfigurasi link Cisco ke Mikrotik menggunakan jaringan 10.10.30.0/30.

```
Router(config)#interface gi0/1
Router(config-if)# description LINK-T0-MIKROTIK
Router(config-if)# ip address 10.10.30.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
```

Gambar 37: Konfigurasi Link Cisco ke Mikrotik

```
interface GigabitEthernet0/1
ip address 10.10.30.1 255.255.255.252
no shutdown
```

11. Melakukan verifikasi interface Mikrotik yang terhubung ke Cisco.

```
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO
[admin@MikroTik] >
```

Gambar 38: Verifikasi Interface Mikrotik Ether2

12. Melakukan verifikasi seluruh interface pada Cisco Router.

```
Router#show ip interface brief
Interface                               IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0                     10.10.11.1      YES manual up            up
GigabitEthernet0/1                     10.10.30.1      YES manual up            up
GigabitEthernet0/2                     unassigned      YES unset up            up
GigabitEthernet0/2.10                  192.168.10.1    YES manual up            up
GigabitEthernet0/2.20                  192.168.20.1    YES manual up            up
GigabitEthernet0/2.30                  192.168.30.1    YES manual up            up
GigabitEthernet0/2.40                  192.168.40.1    YES manual up            up
GigabitEthernet0/3                     10.10.10.1      YES manual up            up
GigabitEthernet0/4                     unassigned      YES unset administratively down down
GigabitEthernet0/5                     unassigned      YES unset administratively down down
GigabitEthernet0/6                     unassigned      YES unset administratively down down
```

Gambar 39: Verifikasi Seluruh Interface Cisco

show ip interface brief

13. Melakukan verifikasi seluruh interface pada Huawei Router.

```
[-HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                               IP Address/Mask  Physical  Protocol  VPN
Ethernet1/0/0                          10.10.10.2/30    up        up        --
Ethernet1/0/0.4094                      128.1.138.137/16 up        up        l3vpn
Ethernet1/0/1                          unassigned       up        down      --
Ethernet1/0/1.4094                      128.1.138.137/16 up        up        l3vpn
Ethernet1/0/2                          192.168.50.1/24  up        up        --
Ethernet1/0/2.4094                      128.1.138.137/16 up        up        l3vpn
Ethernet1/0/3                          10.10.20.1/30    up        up        --
Ethernet1/0/3.4094                      128.1.138.137/16 up        up        l3vpn
Ethernet1/0/4                          10.10.11.2/30    up        up        --
Ethernet1/0/4.4094                      128.1.138.137/16 up        up        l3vpn
---- More ----
```

Gambar 40: Verifikasi Seluruh Interface Huawei

display ip interface brief

14. Melakukan verifikasi seluruh alamat IP pada Mikrotik.

```
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >
```

Gambar 41: Verifikasi Alamat IP Mikrotik

/ip address print

15. Melakukan pengujian konektivitas dari Cisco Router ke Huawei dan Mikrotik.

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/4 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#
```

Gambar 42: Ping Cisco ke Huawei dan Mikrotik

16. Melakukan pengujian konektivitas dari Huawei Router ke Cisco.

```
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=8 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/3/8 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms
```

Gambar 43: Ping Huawei ke Cisco

17. Melakukan pengujian konektivitas dari Huawei Router ke Mikrotik.


```
[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms
```

Gambar 44: Ping Huawei ke Mikrotik

18. Melakukan pengujian konektivitas dari Mikrotik ke Cisco dan Huawei.

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
  0 10.10.20.1                          56 255 6ms
  1 10.10.20.1                          56 255 1ms
  2 10.10.20.1                          56 255 1ms
  3 10.10.20.1                          56 255 1ms
  4 10.10.20.1                          56 255 1ms
  5 10.10.20.1                          56 255 1ms
  6 10.10.20.1                          56 255 1ms
  7 10.10.20.1                          56 255 1ms
  8 10.10.20.1                          56 255 1ms
  9 10.10.20.1                          56 255 1ms
 10 10.10.20.1                          56 255 1ms
 11 10.10.20.1                          56 255 1ms
sent=12 received=12 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=6ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
  0 10.10.30.1                          56 255 2ms
  1 10.10.30.1                          56 255 2ms
  2 10.10.30.1                          56 255 2ms
sent=3 received=3 packet-loss=0% min-rtt=2ms avg-rtt=2ms max-rtt=2ms
```

Gambar 45: Ping Mikrotik ke Cisco dan Huawei

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengkonfigurasi OSPF Cost pada interface Cisco Router.

```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 10.10.10.0 0.0.0.3 area 0
Router(config-router)# network 10.10.11.0 0.0.0.3 area 0
Router(config-router)# network 10.10.30.0 0.0.0.3 area 0
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router(config-router)# network 192.168.30.0 0.0.0.255 area 0
Router(config-router)# network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
```

Gambar 46: Konfigurasi OSPF Cost pada Cisco Router

2. Mengaktifkan OSPF dan mengiklankan seluruh jaringan pada Cisco Router.

```

Router(config)#interface gi0/3
Router(config-if)# description LINK1-TO-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# description LINK2-TO-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/1
Router(config-if)# description LINK-TO-MIKROTIK
Router(config-if)# ip ospf cost 100
Router(config-if)#exit
Router(config)#

```

Gambar 47: Konfigurasi OSPF pada Cisco Router

3. Melakukan verifikasi konfigurasi OSPF pada Cisco Router.

```

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
  --More--

```

Gambar 48: Verifikasi Konfigurasi OSPF Cisco

4. Mengaktifkan OSPF pada Huawei Router.

```

[~HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1] area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0] quit
[*HUAWEI-ospf-1]quit
[*HUAWEI]

```

Gambar 49: Konfigurasi OSPF Huawei Router

5. Mengkonfigurasi OSPF Cost pada interface Huawei Router.

```

[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4] description LINK2-TO-CISCO
[*HUAWEI-Ethernet1/0/4] ospf cost 10
[*HUAWEI-Ethernet1/0/4] quit
[*HUAWEI]
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3] description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ospf cost 100
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]

```

Gambar 50: Konfigurasi OSPF Cost Huawei

6. Melakukan verifikasi neighbor OSPF pada Huawei Router.

```

[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

          OSPF Process 1 with Router ID 2.2.2.2
          Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.10.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1             BDR: 10.10.10.2         MTU: 1500
  Dead timer due in 36 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m31s
  Neighbor Up Time : 2026-06-06 04:16:59
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.11.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1             BDR: 10.10.11.2         MTU: 1500
  Dead timer due in 30 sec
  Retrans timer interval: 5
  ---- More ----

```

Gambar 51: Verifikasi Neighbor OSPF Huawei

Perintah yang digunakan:

```
display ospf peer
```

7. Melakukan verifikasi routing table OSPF pada Huawei Router.

```
-----
_public_ Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
 10.10.30.0/30      OSPF    10   110        D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   110        D   10.10.10.1      Ethernet1/0/0
 192.168.10.0/24    OSPF    10   11          D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   11          D   10.10.10.1      Ethernet1/0/0
 192.168.20.0/24    OSPF    10   11          D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   11          D   10.10.10.1      Ethernet1/0/0
 192.168.30.0/24    OSPF    10   11          D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   11          D   10.10.10.1      Ethernet1/0/0
 192.168.40.0/24    OSPF    10   11          D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   11          D   10.10.10.1      Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
 10.10.10.0/30      OSPF    10   10          10.10.10.2      Ethernet1/0/0
 10.10.11.0/30      OSPF    10   10          10.10.11.2      Ethernet1/0/4
 10.10.20.0/30      OSPF    10  100          10.10.20.1      Ethernet1/0/3
 192.168.50.0/24    OSPF    10   1           192.168.50.1    Ethernet1/0/2

[~HUAWEI]
```

Gambar 52: Verifikasi Routing Table OSPF Huawei

8. Melakukan verifikasi database routing OSPF pada Huawei Router.

```
[~HUAWEI]display ospf routing

      OSPF Process 1 with Router ID 2.2.2.2
      Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
-----
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9  Inter Area: 0  ASE: 0  NSSA: 0
---- More ----
```

Gambar 53: Verifikasi Database Routing OSPF Huawei

9. Mengaktifkan OSPF pada Mikrotik Router.

```

[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=25s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=37s
[admin@MikroTik] >

```

Gambar 54: Konfigurasi OSPF Mikrotik

10. Melakukan verifikasi routing OSPF pada Mikrotik.

```

[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
       - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
       B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 ADo  10.10.10.0/30      10.10.20.1    110
      10.10.30.1
1 ADo  10.10.11.0/30      10.10.20.1    110
      10.10.30.1
2 ADo  192.168.10.0/24     10.10.30.1    110
3 ADo  192.168.20.0/24     10.10.30.1    110
4 ADo  192.168.30.0/24     10.10.30.1    110
5 ADo  192.168.40.0/24     10.10.30.1    110
6 ADo  192.168.50.0/24     10.10.20.1    110
[admin@MikroTik] >

```

Gambar 55: Verifikasi Routing OSPF Mikrotik

11. Melakukan verifikasi neighbor OSPF pada Cisco Router.

```

Router#show ip ospf neighbor
Neighbor ID    Pri  State           Dead Time   Address        Interface
3.3.3.3        1    FULL/BDR        00:00:36    10.10.30.2     GigabitEthernet0/1
2.2.2.2        1    FULL/BDR        00:00:33    10.10.11.2     GigabitEthernet0/0
2.2.2.2        1    FULL/BDR        00:00:36    10.10.10.2     GigabitEthernet0/3
Router#

```

Gambar 56: Verifikasi Neighbor OSPF Cisco

Perintah yang digunakan:

```
show ip ospf neighbor
```

12. Melakukan verifikasi neighbor OSPF secara detail pada Huawei Router.

```

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1          Address: 10.10.10.1
  State: Full              Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1          BDR: 10.10.10.2        MTU: 1500
  Dead timer due in 33 sec
  Retrans timer interval: 5
  Neighbor is up for 00h12m08s
  Neighbor Up Time : 2026-06-06 04:16:59
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3          Address: 10.10.20.2
  State: Full              Mode:Nbr is Master    Priority: 1
  DR: 10.10.20.1          BDR: 10.10.20.2        MTU: 1500
  Dead timer due in 35 sec
  Retrans timer interval: 5
  Neighbor is up for 00h04m56s
  Neighbor Up Time : 2026-06-06 04:24:11
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1          Address: 10.10.11.1
  State: Full              Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1          BDR: 10.10.11.2        MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5
  Neighbor is up for 00h12m08s
  Neighbor Up Time : 2026-06-06 04:16:59
  Authentication Sequence: [ 0 ]

~HUAWEI]

```

Gambar 57: Verifikasi Detail Neighbor Huawei

13. Melakukan verifikasi interface OSPF pada Mikrotik Router.

```

[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=6m16s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=6m28s

```

Gambar 58: Verifikasi Interface OSPF Mikrotik

14. Melakukan verifikasi routing table OSPF pada Cisco Router.

```

Router#show ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1    FULL/BDR        00:00:36   10.10.30.2     GigabitEthernet0/1
2.2.2.2          1    FULL/BDR        00:00:33   10.10.11.2     GigabitEthernet0/0
2.2.2.2          1    FULL/BDR        00:00:36   10.10.10.2     GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:14:30 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:14:30 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:14:30 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1

```

Gambar 59: Verifikasi Routing Table Cisco

Perintah yang digunakan:

show ip route ospf

15. Melakukan verifikasi database OSPF pada Cisco Router.

```

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:15:12, GigabitEthernet0/0
      [110/110] via 10.10.10.2, 00:15:12, GigabitEthernet0/3
O    192.168.50.0/24 [110/11] via 10.10.11.2, 00:15:12, GigabitEthernet0/0
      [110/11] via 10.10.10.2, 00:15:12, GigabitEthernet0/3

```

Gambar 60: Verifikasi Database OSPF Cisco

16. Melakukan verifikasi routing table pada Huawei Router.


```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole r
-----
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost      Flags NextHop         Interface
-----
10.10.30.0/30       OSPF   10   110       D    10.10.11.1        Ethernet1/0/4
10.10.10.0/30       OSPF   10   110       D    10.10.10.1        Ethernet1/0/0
192.168.10.0/24     OSPF   10   11         D    10.10.11.1        Ethernet1/0/4
192.168.20.0/24     OSPF   10   11         D    10.10.10.1        Ethernet1/0/0
192.168.20.0/24     OSPF   10   11         D    10.10.11.1        Ethernet1/0/4
192.168.30.0/24     OSPF   10   11         D    10.10.10.1        Ethernet1/0/0
192.168.30.0/24     OSPF   10   11         D    10.10.11.1        Ethernet1/0/4
192.168.40.0/24     OSPF   10   11         D    10.10.10.1        Ethernet1/0/0
192.168.40.0/24     OSPF   10   11         D    10.10.11.1        Ethernet1/0/4
192.168.40.0/24     OSPF   10   11         D    10.10.10.1        Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto  Pre  Cost      Flags NextHop         Interface
-----
10.10.10.0/30       OSPF   10   10         10.10.10.2        Ethernet1/0/0
10.10.11.0/30       OSPF   10   10         10.10.11.2        Ethernet1/0/4
10.10.20.0/30       OSPF   10   100        10.10.20.1        Ethernet1/0/3
192.168.50.0/24     OSPF   10   1          192.168.50.1      Ethernet1/0/2
[~HUAWEI]

```

Gambar 61: Verifikasi Routing Table Huawei

17. Melakukan verifikasi routing table pada Mikrotik Router.

```

[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo  10.10.10.0/30      10.10.20.1 110
      10.10.30.1
1 ADo  10.10.11.0/30      10.10.20.1 110
      10.10.30.1
2 ADo  192.168.10.0/24    10.10.30.1 110
3 ADo  192.168.20.0/24    10.10.30.1 110
4 ADo  192.168.30.0/24    10.10.30.1 110
5 ADo  192.168.40.0/24    10.10.30.1 110
6 ADo  192.168.50.0/24    10.10.20.1 110

```

Gambar 62: Verifikasi Routing Table Mikrotik

18. Melakukan pengujian konektivitas dari LAN 50 menuju seluruh VLAN pada Cisco Router.

```

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.923 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=9.919 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.368 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.184 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.126 ms

```

Gambar 63: Pengujian Konektivitas LAN 50

19. Melakukan pengujian konektivitas menuju VLAN 10.

```
VPCS> ping 192.168.50.10  
  
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.961 ms  
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.975 ms  
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.738 ms  
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.690 ms  
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.174 ms
```

Gambar 64: Ping Menuju VLAN 10

20. Melakukan pengujian konektivitas menuju VLAN 20.

```
VPCS> ping 192.168.50.10  
  
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.699 ms  
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.453 ms  
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.043 ms  
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.226 ms  
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.430 ms
```

Gambar 65: Ping Menuju VLAN 20

21. Melakukan pengujian konektivitas menuju VLAN 30.

```
VPCS> ping 192.168.50.10  
  
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=8.122 ms  
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.835 ms  
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.895 ms  
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.932 ms  
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.163 ms
```

Gambar 66: Ping Menuju VLAN 30

22. Melakukan pengujian konektivitas menuju VLAN 40.

```

VPCS> ping 192.168.10.11
64 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=14.175 ms
64 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=6.334 ms
64 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=7.382 ms
64 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=6.997 ms
64 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=6.109 ms

VPCS> ping 192.168.20.11
64 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=8.639 ms
64 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=8.495 ms
64 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=8.954 ms
64 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=7.646 ms
64 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=9.236 ms

```

Gambar 67: Ping Menuju VLAN 40

23. Melakukan pengujian konektivitas antar jaringan setelah seluruh route OSPF terbentuk.

```

VPCS> ping 192.168.30.10
64 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=7.972 ms
64 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=8.237 ms
64 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=8.348 ms
64 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=8.652 ms
64 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=8.564 ms
64 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=8.564 ms

VPCS> ping 192.168.40.10
64 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=16.290 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=7.456 ms
64 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=9.769 ms
64 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=7.658 ms
64 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=9.808 ms

```

Gambar 68: Pengujian Konektivitas Akhir OSPF

1.5 Praktikum 5: Pengujian Failover OSPF

1. Melakukan pengujian jalur awal menuju jaringan LAN 50 menggunakan traceroute.

```

VPCS> trace 192.168.50.10
traceroute to 192.168.50.10, 8 hops max
 1 192.168.50.10      0.001 ms

```

Gambar 69: Traceroute Sebelum Failover

Perintah yang digunakan:

```
trace 192.168.50.10
```

2. Memutus jalur utama OSPF dengan menonaktifkan interface GigabitEthernet0/3 pada Cisco Router.

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 04:51:32.137: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down
: Interface down or detached
*Jun 6 04:51:34.185: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun 6 04:51:35.186: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
```

Gambar 70: Menonaktifkan Interface Jalur Utama

Perintah yang digunakan:

```
interface GigabitEthernet0/3
shutdown
```

3. Melakukan verifikasi route OSPF setelah jalur utama dimatikan.

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:01:28 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:01:28 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
```

Gambar 71: Verifikasi Route OSPF Setelah Failover

Perintah yang digunakan:

```
show ip route 192.168.50.0
```

4. Melakukan pengujian traceroute setelah perubahan jalur routing.

```
/PCS> trace 192.168.50.10
tracert to 192.168.50.10, 8 hops max
 1 192.168.50.10      0.001 ms
```

Gambar 72: Traceroute Setelah Failover

Perintah yang digunakan:

```
trace 192.168.50.10
```

5. Melakukan verifikasi route OSPF setelah konvergensi jaringan.

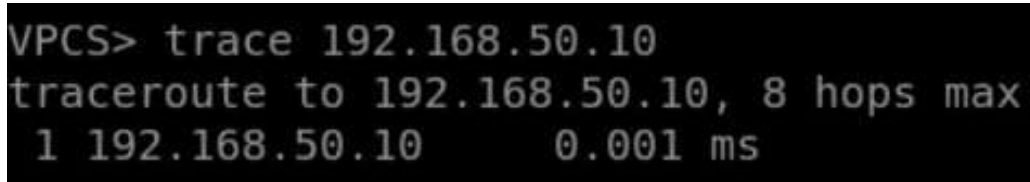
```
Router#show ip route 192.168.50.10
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.10.2 on GigabitEthernet0/3, 00:32:52 ago
  Routing Descriptor Blocks:
    * 10.10.10.2, from 2.2.2.2, 00:32:52 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
```

Gambar 73: Verifikasi Route OSPF Setelah Konvergensi

Perintah yang digunakan:

```
show ip route 192.168.50.0
```

6. Melakukan pengujian traceroute kembali untuk melihat jalur yang digunakan setelah konvergensi.



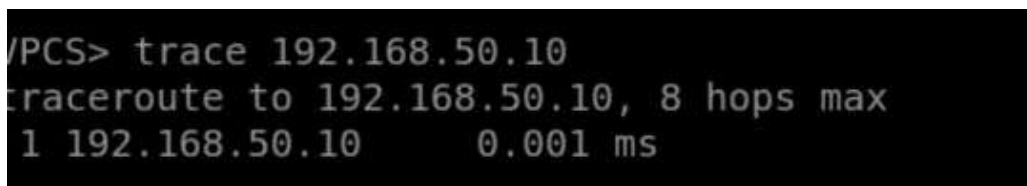
```
VPCS> trace 192.168.50.10
tracert to 192.168.50.10, 8 hops max
 1 192.168.50.10      0.001 ms
```

Gambar 74: Traceroute Setelah Konvergensi OSPF

Perintah yang digunakan:

```
trace 192.168.50.10
```

7. Melakukan pengujian akhir konektivitas jaringan setelah proses failover.



```
VPCS> trace 192.168.50.10
tracert to 192.168.50.10, 8 hops max
 1 192.168.50.10      0.001 ms
```

Gambar 75: Pengujian Akhir Failover OSPF

Perintah yang digunakan:

```
trace 192.168.50.10
```

2 Analisis Hasil Percobaan

Pada Modul 5 ini dilakukan implementasi jaringan multivendor yang terdiri dari Cisco Switch, Cisco Router, Huawei Router, dan Mikrotik Router. Tahap awal praktikum dimulai dengan konfigurasi VLAN dan trunking pada Cisco Switch. Hasil verifikasi menunjukkan bahwa VLAN 10, VLAN 20, VLAN 30, dan VLAN 40 berhasil dibuat serta dapat dilewatkan melalui trunk menggunakan protokol IEEE 802.1Q. Konfigurasi access port dan trunk port juga berjalan sesuai dengan topologi yang telah dirancang.

Tahap berikutnya adalah konfigurasi Router-on-a-Stick pada Cisco Router menggunakan subinterface untuk setiap VLAN. Selain itu, DHCP Server dikonfigurasi untuk VLAN 10 dan VLAN 20 sehingga host dapat memperoleh alamat IP secara otomatis. Hasil pengujian menunjukkan bahwa perangkat berhasil memperoleh alamat IP dari DHCP Server dan dapat melakukan komunikasi dengan gateway maupun antar VLAN. Hal ini menunjukkan bahwa proses inter-VLAN routing telah berjalan dengan baik.

Selanjutnya dilakukan konfigurasi alamat IP pada jaringan antar router yang menghubungkan Cisco Router, Huawei Router, dan Mikrotik Router. Pengujian konektivitas menggunakan perintah ping menunjukkan bahwa seluruh router dapat saling berkomunikasi sehingga konfigurasi IP addressing dan koneksi antar perangkat berhasil dilakukan.

Pada implementasi routing dinamis menggunakan protokol OSPF, setiap router berhasil membentuk hubungan neighbor dan melakukan pertukaran informasi routing. Hal ini dibuktikan dengan munculnya neighbor OSPF pada masing-masing perangkat serta bertambahnya route OSPF pada tabel routing Cisco, Huawei, dan Mikrotik. Selain itu, jaringan yang sebelumnya tidak terhubung secara langsung dapat saling berkomunikasi melalui route yang dipelajari dari OSPF.

Namun, pada tahap pengujian failover OSPF ditemukan bahwa perpindahan jalur belum berjalan sesuai dengan yang diharapkan. Ketika salah satu link dimatikan, proses pemilihan jalur alternatif tidak selalu berlangsung dengan baik sehingga beberapa pengujian konektivitas menunjukkan hasil yang kurang sesuai dengan tujuan praktikum. Kondisi ini dapat disebabkan oleh konfigurasi cost yang kurang tepat, proses konvergensi yang belum optimal, atau adanya konfigurasi routing yang belum sepenuhnya sinkron pada seluruh perangkat.

Secara keseluruhan, konfigurasi VLAN, trunking, inter-VLAN routing, DHCP Server, konektivitas antar router, dan OSPF multivendor berhasil dilakukan. Kendala utama yang ditemukan berada pada mekanisme failover OSPF yang masih memerlukan perbaikan agar proses redundansi jaringan dapat berjalan secara optimal.

3 Hasil Tugas Modul

Hasil pengerjaan tugas modul dapat dilihat di README.md folder Modul 5/LA di github repository kelompok 2

4 Kesimpulan

Berdasarkan hasil praktikum Modul 5 yang telah dilakukan, konfigurasi VLAN dan trunking pada Cisco Switch berhasil diterapkan sehingga segmentasi jaringan dapat dilakukan dengan baik. Implementasi Router-on-a-Stick pada Cisco Router juga berhasil memungkinkan komunikasi antar VLAN, sedangkan DHCP Server berhasil memberikan alamat IP secara otomatis kepada host sesuai dengan jaringan VLAN masing-masing.

Konfigurasi alamat IP antar router Cisco, Huawei, dan Mikrotik berhasil membentuk konektivitas jaringan antar perangkat. Selain itu, implementasi routing dinamis menggunakan protokol OSPF berhasil membentuk hubungan neighbor dan mendistribusikan informasi routing sehingga jaringan yang berbeda dapat saling berkomunikasi tanpa menggunakan route statis.

Meskipun konfigurasi dasar OSPF berhasil dilakukan, pengujian failover belum memberikan hasil yang sepenuhnya sesuai dengan harapan. Perpindahan jalur saat terjadi gangguan pada salah satu link masih memerlukan penyempurnaan konfigurasi agar mekanisme redundansi dapat berjalan secara optimal.

Dengan demikian, tujuan utama praktikum telah berhasil dicapai, yaitu memahami implementasi VLAN, trunking, inter-VLAN routing, DHCP Server, konektivitas antar router, dan routing dinamis OSPF pada lingkungan multivendor. Kendala yang ditemukan pada pengujian failover menjadi bahan evaluasi untuk meningkatkan keandalan jaringan pada implementasi berikutnya.

5 Lampiran



Gambar 76: Dokumentasi Praktikum 1